

Random Variables:

S - Set of possible outcomes.

$$S = \{H, T\}, \quad S = \{HH, HT, TH, TT\}.$$

A random variable is a function $X: S \rightarrow \mathbb{R}$.

eg: $S: \{H, T\}$

$$X: S \rightarrow \mathbb{R}, \text{ define } X(H) = 1, X(T) = -1$$

$$X: S \rightarrow \mathbb{R}, \text{ define } X(H) = \frac{1}{2}, X(T) = \frac{1}{2}$$

eg: $S = \{HH, HT, TH, TT\}$

$$X(HH) = 0, X(HT) = 1, X(TH) = -1, X(TT) = 2$$

$$X(HH) = \frac{1}{4}, X(HT) = \frac{1}{4}, X(TH) = \frac{1}{4}, X(TT) = \frac{1}{4}$$

Definition 6. A probability of an event A is the sum of the weights of all sample points in A such that $P(S) = 1, P(A) \geq 0, P(\emptyset) = 0$

Example 5. A coin is tossed twice. What is the probability that at least 1 head occurs?

X = tossing a coin twice

Next Gen. - $\{HH, HT, TH, TT\}$.

$$P(X=0) = \frac{1}{4}, P(X=1) = \frac{2}{4}, P(X=2) = \frac{1}{4}$$

$$P(X \geq 1) = ?$$

$$P(X=0) + P(X=1) + P(X=2) = 1$$

$$P(X \geq 1) = \frac{2}{4} + \frac{1}{4} = \frac{3}{4}$$

Theorem 1. If A and B are two events, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

• $P(A \cup B) = P(A) + P(B)$ if A and B are mutually exclusive

• $P(A') = 1 - P(A)$ $P(A \cup A') = 1 \Rightarrow P(A) + P(A') = 1$.

Example 6. What is the probability of getting a total 7 or 11 when a pair of fair dice is tossed?

$$\begin{aligned} P(\text{getting a total 7 or 11}) &= P(\text{getting a total 7}) \\ &\quad + P(\text{getting a total 11}) \\ &= P((1,6), (2,5), (3,4), (4,3), (5,2), (6,1), (1,6)) \\ &\quad + P((5,6), (6,5)) \\ &= \frac{8}{36} \end{aligned}$$

Example 7. Two balls are drawn in succession without replacement from an urn containing 4 red balls and 3 black balls. The possible outcomes and the values y of the random variable Y , Y is number of red balls.

① Discrete random variable.

Eg: Tossing a coin or rolling a dice.

② Continuous random variable.

Eg: Problems related to Temp or wave freq.

Probability mass fun: (PMF)

A function $f: X \rightarrow \mathbb{R}$ is said to be pmf if

$\exists$ Prob. fun $\exists f(x) = P(X=x)$ it satisfies

① $P(X=x) \geq 0$ ② $\sum_{x \in X} P(X=x) = 1$ (Discrete R.V.)

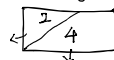
② $P(X=x) \geq 0$ ② $\int_{-\infty}^{\infty} P(X=x) dx = 1$ (Continuous R.V.)

Example 8. A shipment of 6 television sets contains 2 defective sets. A hotel makes a random purchase 3 of the sets. If X is number of defective sets purchased by hotel. Find the probability distribution.

$$\begin{aligned} X &= \text{No. of defective sets purchased by the hotel.} \\ &= 0, 1, 2 \end{aligned}$$

$$P(X=0) = \frac{{}^4C_3}{{}^6C_3} = \frac{4!}{3!(4-3)!} \times \frac{3!(6-3)!}{6!} = \frac{2 \times 3}{6 \times 6} = \frac{1}{5}$$

$$P(X=1) = \frac{{}^2C_1 {}^4C_2}{{}^6C_3} = \frac{2!}{1! \times 1!} \times \frac{4!}{2! \times 2!} \times \frac{3!(6-3)!}{6!} = \frac{3 \times 2 \times 2}{6 \times 6} = \frac{2}{5}$$



$$P(X=0) = \frac{{}^6C_3}{{}^6C_3} = \frac{4!}{3!(4-3)!} = \frac{2 \times 3}{6 \times 6} = \frac{1}{5}$$

$$P(X=1) = \frac{{}^6C_1 {}^5C_2}{{}^6C_3} = \frac{6!}{1! \times 1! \times 2! \times 2!} \times \frac{3! (6-3)!}{6!} = \frac{3 \times 2 \times 2}{5 \times 6} = \frac{2}{5}$$

$$P(X=2) = \frac{{}^6C_2 {}^4C_1}{{}^6C_3} = \frac{4!}{2! \times 2!} \times \frac{3! (6-3)!}{6!} = \frac{2 \times 3}{5 \times 6} = \frac{1}{5}$$

The corresponding Pdf is

x	0	1	2
$P(X=x)$	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{1}{5}$

Example 9. A random variable X assume 4 values with probabilities

$\frac{1+3x}{4}, \frac{1-x}{4}, \frac{1+2x}{4}, \frac{1-4x}{4}$. Find condition on x , such that these values represent the probability function of x .

$$p_0 = \frac{1+3x}{4}, p_1 = \frac{1-x}{4}, p_2 = \frac{1+2x}{4}, p_3 = \frac{1-4x}{4}$$

① $p_i \geq 0 \forall i$

$$p_0 \geq 0 \Rightarrow \frac{1+3x}{4} \geq 0 \Rightarrow 1+3x \geq 0 \Rightarrow 3x \geq -1 \Rightarrow x \geq -\frac{1}{3}$$

$$p_1 \geq 0 \Rightarrow \frac{1-x}{4} \geq 0 \Rightarrow 1-x \geq 0 \Rightarrow x \leq 1$$

$$p_2 \geq 0 \Rightarrow \frac{1+2x}{4} \geq 0 \Rightarrow 1+2x \geq 0 \Rightarrow 2x \geq -1 \Rightarrow x \geq -\frac{1}{2}$$

$$p_3 \geq 0 \Rightarrow \frac{1-4x}{4} \geq 0 \Rightarrow 1-4x \geq 0 \Rightarrow x \leq \frac{1}{4}$$

for $x \in \left[-\frac{1}{3}, \frac{1}{4}\right]$, $p_i \geq 0$

$$p_0 + p_1 + p_2 + p_3 = 1$$

$\Rightarrow$ for $x \in \left[-\frac{1}{3}, \frac{1}{4}\right]$, the given r.v represents prob. fun.

eg:

Example 10. If the random variable X takes the values of 1, 2, 3, 4 such that $2P(X=1) = 3P(X=2) = P(X=3) = 5P(X=4)$. Find the probability distribution function of X .

Given $2P(X=1) = 3P(X=2) = P(X=3) = 5P(X=4) = k$

$$P(X=1) = \frac{k}{2}, P(X=2) = \frac{k}{3}, P(X=3) = k, P(X=4) = \frac{k}{5}$$

$$P(X=1) + P(X=2) + P(X=3) + P(X=4) = 1$$

$$\frac{k}{2} + \frac{k}{3} + k + \frac{k}{5} = 1$$

$$\frac{5k}{6} + \frac{6k}{5} = 1$$

$$\frac{(25+36)k}{30} = 1 \Rightarrow \frac{61}{30}k = 1$$

$$k = \frac{30}{61}$$

x	1	2	3	4
$P(X=x)$	$\frac{15}{61}$	$\frac{10}{61}$	$\frac{30}{61}$	$\frac{6}{61}$

Example 12. Two cards are drawn successively with replacement from a well shuffled pack of 52 cards. Find the probability distribution of the number of kings.

Example 13. Find the probability distribution of the number of sixes in three tosses of a dice.

Example 14. A urn contains 4 white and 3 red balls. Find the probability distribution of the red balls in three draws with replacement, from the urn. Also find its mean and variance.

$$P(X=x)$$

$$\text{Cumulative distribution function: } F(x) = P(X \leq x) = \sum_{t \leq x} P(X=t)$$

Ex:

Cumulative distribution function: $F(x) = P(X \leq x) = \sum_{t \leq x} P(X \leq t)$.

Ex: Find the cumulative dist. function for pdf

x	0	1	2	3
$P(X=x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$	0

$$P(X=0) = \frac{1}{4}, \quad P(X \leq 1) = P(X=0) + P(X=1) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

$$P(X \leq 2) = \frac{3}{4} + \frac{1}{4} = 1$$

$$P(X \leq 3) = 1 + 0 = 1$$

$$P(X=x), \quad \text{Mean} = E(X) = \sum_{x \in X} x P(X=x)$$

$$E(X^2) = \sum_{x \in X} x^2 P(X=x)$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Example 10. A random variable X has the following probability distribution

x	0	1	2	3	4	5	6	7
$P(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

- Find k
- $P\left(\frac{1.5 < X < 4.5}{X > 2}\right)$
- The smallest value of λ for which $P(X \leq \lambda) > \frac{1}{2}$

Example 11. The probability of an infinite discrete distribution is given by $P(X = j) = \frac{1}{2^j}, j = 1, 2, \dots$. Verify that the total probability is 1 and find mean and variance of the distribution. Find

- $P(X \text{ is even})$,
- $P(X \geq 5)$,
- $P(X \text{ is divisible by } 3)$

Example 12. If $P(X = r) = k(1-a)^{r-1}, 0 < a < 1$ where $r = 1, 2, 3, \dots$ find k , mean and variance of X

$$\sum_{r=1}^{\infty} P(X=r) = 1 \Rightarrow \sum_{r=1}^{\infty} k(1-a)^{r-1} = 1$$

$$k \sum_{r=1}^{\infty} (1-a)^{r-1} = 1 \quad \sum_{r=0}^{\infty} x^r = \frac{1}{1-x}, \quad x < 1$$

$$x = 1-a$$

$$k \left[(1-a)^0 + (1-a) + (1-a)^2 + (1-a)^3 + \dots \right] = 1$$

$$k \left[\frac{1}{1-(1-a)} \right] = 1 \Rightarrow \frac{k}{a} = 1 \Rightarrow \boxed{k=a}$$

for $k=a$, $P(X=r) = a(1-a)^{r-1}, r=1, 2, 3, \dots$ is pdf.

$$\text{Mean} = E(X) = \sum_{r=1}^{\infty} r P(X=r) = \sum_{r=1}^{\infty} r a(1-a)^{r-1} = a \sum_{r=1}^{\infty} r (1-a)^{r-1}$$

$$\text{Let } x = 1-a$$

$$\sum_{r=1}^{\infty} r (1-a)^{r-1} = \sum_{r=1}^{\infty} r x^{r-1} = 1 + 2x + 3x^2 + 4x^3 + \dots$$

$$= \frac{1}{(1-x)^2} = \frac{1}{(1-(1-a))^2} = \frac{1}{a^2}$$

$$\text{Mean} = E(X) = a \sum_{r=1}^{\infty} r (1-a)^{r-1} = a \cdot \frac{1}{a^2} = \frac{1}{a}$$

Example 13. A continuous random variable X has a pdf $f(x) = kx^2e^{-x}, x \geq 0$. Find k , mean and variance.

① $f(x) \geq 0$, $\int_{-\infty}^{\infty} f(x) dx = 1$.

$$\int_0^{\infty} f(x) dx = 1 \Rightarrow \int_0^{\infty} kx^2 e^{-x} dx = 1$$

$$k \int_0^{\infty} x^2 e^{-x} dx = 1$$

$$k \Gamma_3 = 1$$

$$k \cdot 2 = 1 \Rightarrow \boxed{k = \frac{1}{2}}$$

$$\Gamma_n = \int_0^{\infty} x^{n-1} e^{-x} dx \quad \checkmark$$

$$\Gamma_3 = \int_0^{\infty} x^2 e^{-x} dx, \quad n-1=2, \quad n=3$$

$$\Gamma_n = (n-1)!$$

$$\Gamma_3 = (3-1)! = 2! = 2$$

for $k = \frac{1}{2}$, given f is pdf.

$$\begin{aligned} \text{Mean: } E(X) &= \int_0^{\infty} x f(x) dx = \int_0^{\infty} x \cdot \frac{1}{2} x^2 e^{-x} dx = \frac{1}{2} \int_0^{\infty} x^3 e^{-x} dx, \quad n-1=3, \quad n=4 \\ &= \frac{1}{2} \Gamma_4 = \frac{3!}{2} = \frac{1 \times 2 \times 3}{2} \end{aligned}$$

$$\boxed{E(X) = 3}$$

$$\begin{aligned} E(X^2) &= \int_0^{\infty} x^2 f(x) dx = \int_0^{\infty} x^2 \cdot \frac{1}{2} x^2 e^{-x} dx = \frac{1}{2} \int_0^{\infty} x^4 e^{-x} dx, \quad n-1=4, \quad n=5 \\ &= \frac{1}{2} \Gamma_5 = \frac{4!}{2} = 3 \times 4 = 12 \end{aligned}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 12 - (3)^2 = 3$$

$$\boxed{\text{Var } X = 3}$$

Example 14. The distribution function of a random variable X is given by

$F(x) = 1 - (1+x)e^{-x}, x \geq 0$. find the density function, mean and variance of X

f - pdf, $F(x) = \int_{-\infty}^x f(t) dt$ - C.D.F

$$F'(x) = f(x)$$

given, $F(x) = 1 - (1+x)e^{-x}$

$$\begin{aligned} F'(x) &= 0 - e^{-x} + (1+x)e^{-x} \\ &= -e^{-x} + e^{-x} + x e^{-x} \end{aligned}$$

$$F'(x) = x e^{-x} \Rightarrow f(x) = x e^{-x} \text{ is pdf.}$$

$$\text{Mean} = \int_0^{\infty} x f(x) dx = \int_0^{\infty} x^2 e^{-x} dx = \Gamma_3 = 2$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 6 - 2^2 = 2$$

Example: Given C.D.F

$$F(x) = \begin{cases} 0 & x < 0 \\ x^2 & 0 < x < \frac{1}{2} \\ 1 - \frac{3}{25}(3-x)^2 & \frac{1}{2} \leq x < 3 \\ 1 & x \geq 3 \end{cases} \quad \text{Find the PDF of } X \text{ and evaluate } P(|X| \leq 1) \text{ and}$$

$P(\frac{1}{3} \leq X < 4)$ using both PDF and CDF

① Note $f(x) = F'(x) = \begin{cases} 0 & x < 0 \\ 2x & 0 < x < \frac{1}{2} \\ +\frac{6}{5}(3-x) & \frac{1}{2} \leq x < 3 \\ 0 & x \geq 3 \end{cases}$

$$f(x) = \begin{cases} 2x & 0 < x < \frac{1}{2} \\ \frac{6}{25}(3-x) & \frac{1}{2} \leq x < 3 \\ 0 & \text{otherwise} \end{cases} \text{ is pdf.}$$

$$\begin{aligned} \textcircled{2} \quad P(|x| < 1) &= P(-1 < x < 1) = \int_{-1}^1 f(x) dx = \int_0^1 f(x) dx \\ &= \int_0^{\frac{1}{2}} f(x) dx + \int_{\frac{1}{2}}^1 f(x) dx \\ &= \int_0^{\frac{1}{2}} 2x dx + \int_{\frac{1}{2}}^1 \frac{6}{25}(3-x) dx \\ &= 2x^2 \Big|_0^{\frac{1}{2}} + \frac{6}{25} \left(3x - \frac{x^2}{2} \right) \Big|_{\frac{1}{2}}^1 = \frac{1}{4} + \frac{6}{25} \left(3 - \frac{1}{2} - \frac{3}{2} + \frac{1}{8} \right) \end{aligned}$$

$\textcircled{3}$ Find $P(\frac{1}{3} < x < 4)$ using cdf & pdf.

$$\begin{aligned} P(a < x < b) &= \int_a^b f(x) dx = \int_a^b F'(x) dx = F(b) - F(a) \\ P(\frac{1}{3} < x < 4) &= F(4) - F(\frac{1}{3}) = 1 - \left(\frac{1}{3}\right)^2 \\ &= 1 - \frac{1}{9} = \frac{8}{9} \end{aligned}$$

Example 16. If $p(x) = \begin{cases} xe^{-\frac{x^2}{2}} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$

- Show that $p(x)$ is a PDF of a continuous random variable
- Find its distribution function

Example 17. A continuous random variable X that can assume any value between $x = 2$ and $x = 5$ has density function given by $f(x) = k(1+x)$. Find $P(X < 4)$

Example 18. A continuous random variable has a PDF $f(x) = 3x^2, 0 \leq x \leq 1$. Find a and b such that

- $P(X \leq a) = P(X > a)$
- $P(X > b) = 0.05$

Soln: $\textcircled{1} P(x \leq a) = P(x > a)$

$$\begin{aligned} \int_{-\infty}^a f(x) dx &= \int_a^{\infty} f(x) dx \Rightarrow \int_0^a 3x^2 dx = \int_a^1 3x^2 dx \\ 3x^2 \Big|_0^a &= 3x^2 \Big|_a^1 \\ a^3 &= 1 - a^3 \Rightarrow 2a^3 = 1 \\ a &= \frac{1}{\sqrt[3]{2}} \end{aligned}$$

$\textcircled{2} P(x > b) = 0.05$

$$\begin{aligned} \int_b^1 f(x) dx &= 0.05 \Rightarrow \int_b^1 3x^2 dx = 0.05 \\ x^3 \Big|_b^1 &= 0.05 \\ 1 - b^3 &= 0.05 \\ b^3 &= 0.95 \\ b &= (0.95)^{\frac{1}{3}} \end{aligned}$$

$$b^3 = 0.95$$

$$b = (0.95)^{1/3}$$

Example 19. A random variable X has a CDF of the form $F(x) = 1 - (\frac{1}{2})^{x+1}, x = 0, 1, 2, \dots$

- Find the probability function of X
- Find $P(0 < X \leq 8)$.

Example 20. A continuous random variable X has a density function $f(x) = \begin{cases} \frac{2x}{9} & 0 < x < 3 \\ 0 & \text{otherwise} \end{cases}$

- Develop CDF of X
- Find mean and Variance of X
- Find $\mu_3' = E(X^3)$.
- Find the value of m such that $P(X \geq m) = P(X \leq m)$.

$$f(x) = x \quad f(x, y) = xy \quad \left. \begin{array}{l} f(x) = P(X=x) \geq 0 \\ \int_{-\infty}^{\infty} f(x) dx = 1 \end{array} \right\} \begin{array}{l} P(X=j) \geq 0 \\ \sum_j P(X=j) = 1 \end{array}$$

$$X: A \rightarrow \mathbb{R} - \text{R.V.}, \quad Y: B \rightarrow \mathbb{R}.$$

$$X \times Y: A \times B \rightarrow \mathbb{R}. \rightarrow \text{Joint Prob. dist.}$$

Let x & y be a R.V., The joint prob. dist.

$$p: X \times Y \rightarrow \mathbb{R} \ni \exists \text{ pmf } \ni p(i, j) = P(X=i, Y=j).$$

discrete: ① $P(i, j) \geq 0$ ② $\sum_i \sum_j P(X=i, Y=j) = 1$

continuous: $\exists$ function $f \ni f(x, y) = P(X=x, Y=y) \ni$

$$\text{① } f(x, y) \geq 0 \quad \text{② } \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

x	0	1
$P(X=x)$	$\frac{1}{2}$	$\frac{1}{2}$

x - not of heads occurs
 y - depend upon the time.

$y \backslash x$	0	1
0	$\frac{1}{4}$	$\frac{1}{2}$
①	$\frac{1}{2}$	$\frac{1}{4}$
2	0	1

$$\frac{1}{4} + \frac{1}{2} + \frac{1}{2} + \frac{1}{4} + 0 + 1 \neq 1$$

Example 25. For the bivariate probability distribution of (x, y) given below,

$X \backslash Y$	1	2	3	4	5	6	
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$	$\frac{8}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{5}{8}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$	$\frac{4}{32}$
find	$\frac{3}{32}$	$\frac{3}{32}$					1

$$P(X=0, Y=1) = 0$$

$$P(X=0, Y=2) = 0$$

$$P(X=0, Y=3) = \frac{1}{32}$$

$$\text{① } P(X=x, Y=y) \geq 0$$

$$\text{② } \sum_x \sum_y P(X=x, Y=y) = 1$$

$$0 + 0 + \frac{1}{32} + \frac{2}{32} + \frac{2}{32} + \frac{3}{32} + \dots = 1$$

$$X, Y \rightarrow X \times Y$$

$$\bullet P(X \leq 1)$$

$$\bullet P(Y \leq 3)$$

$$P(Y \leq 3)$$

$$\sum_x \sum_y P(X=x, Y=y) = 1$$

$$0 + 0 + \frac{1}{8} + \frac{2}{8} + \dots = 1 \quad x, y \rightarrow x \times y$$

$$f(x, y) = \sin(x+y). \quad y=1, f(x, 1) = \sin(x+1) \quad \text{given } x \times y \rightarrow \frac{x}{y}$$

Marginal distribution: given joint prob dist fun on $x \times y$,
define Marginal dist fun on x ,

$$g(x) = \sum_{y \in Y} P(X=x, Y=y)$$

Marginal dist fun on y ,

$$h(y) = \sum_{x \in X} P(X=x, Y=y)$$

Ex:

$$\frac{1}{2} + \frac{1}{4} = \frac{1}{2}(1 + \frac{1}{2})$$

		*	
x \ y	0	1	
0	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$
1	$\frac{1}{4}$	0	$\frac{1}{4}$
	$\frac{3}{4}$	$\frac{1}{4}$	1

The given dist is pdf.

Marginal dist on x .

$$g(0) = \frac{3}{4}, g(1) = \frac{1}{4}$$

x	0	1
g(x)	$\frac{3}{4}$	$\frac{1}{4}$

Marginal dist on y .

$$h(0) = \frac{3}{4}, h(1) = \frac{1}{4}$$

y	0	1
h(y)	$\frac{3}{4}$	$\frac{1}{4}$

$$\begin{aligned} \textcircled{1} \quad \text{Find } P(X \leq 0, Y \leq 1) &= P(X=0, Y=0) + P(X=0, Y=1) \\ &= \frac{1}{2} + \frac{1}{4} = \frac{3}{4} \end{aligned}$$

$$\textcircled{2} \quad \text{Find } P(X \leq 1) = g(0) + g(1) = \frac{3}{4} + \frac{1}{4} = 1$$

$$\textcircled{3} \quad \text{Find } P(Y \leq 0) = h(0) = \frac{3}{4}$$

$$\textcircled{4} \quad P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\begin{aligned} P(X \leq 1 | Y=0) &= \frac{P(X \leq 1, Y=0)}{P(Y=0)} = \frac{P(X=0, Y=0) + P(X=1, Y=0)}{h(0)} \\ &= \frac{\frac{1}{2} + \frac{1}{4}}{\frac{3}{4}} = 1 \end{aligned}$$

Example 25. For the bivariate probability distribution of (x, y) given below,

$X \backslash Y$	1	2	3	4	5	6
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$

find

- $P(X \leq 1)$
- $P(Y \leq 3)$
- $P(X \leq 1, Y \leq 3)$
- $P(X \leq 1 | Y \leq 3), P(Y \leq 3 | X \leq 1)$
- $P(X \leq 1 | Y \leq 3), P(Y \leq 3 | X \leq 1)$

Example 24. A privately owned business operates both a drive in facility and walk in facility. On a randomly selected day, Let X and Y be the proportions of the time that the drive in and walk in facilities are in use. Let $f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y) & 0 \leq x, y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$

- Verify f is joint PDF
- Find $P([X, Y] \in A)$ where $A = \{(x, y) | 0 < x < \frac{1}{2}, \frac{1}{4} < y < \frac{1}{2}\}$

$$\begin{aligned} \textcircled{1} \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy &= 1 \Rightarrow \int_0^1 \int_0^1 \frac{2}{5} (2x + 3y) dx dy = \int_0^1 \left[\frac{2}{5} \left(\frac{2x^2}{2} + 3yx \right) \right]_0^1 dy \\ &= \int_0^1 \frac{2}{5} [1 + 3y] dy \\ &= \frac{2}{5} \left[y + 3y^2/2 \right]_0^1 = \frac{2}{5} \left[1 + \frac{3}{2} \right] = 1 \end{aligned}$$

$\Rightarrow$ given f is joint PDF.

$$\begin{aligned} \textcircled{2} \quad \text{Marginal dist on } x &= g(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^1 \frac{2}{5} (2x + 3y) dy \\ &= \frac{2}{5} \left[2xy + 3y^2/2 \right]_0^1 = \frac{2}{5} \left[2x + \frac{3}{2} \right] \end{aligned}$$

$$g(x) = \frac{2}{5} \left[2x + \frac{3}{2} \right]$$

$$\begin{aligned} \textcircled{3} \quad \text{Marginal dist on } y, \quad h(y) &= \int_{-\infty}^{\infty} f(x, y) dx = \int_0^1 \frac{2}{5} (2x + 3y) dx \\ &= \frac{2}{5} \left[2x^2/2 + 3yx \right]_0^1 \\ &= \frac{2}{5} [1 + 3y] \end{aligned}$$

$$h(y) = \frac{2}{5} [1 + 3y]$$

$$\textcircled{4} \quad P\left\{ (x, y) \mid 0 \leq x \leq \frac{1}{2}, \frac{1}{4} \leq y \leq \frac{1}{2} \right\} = \int_{\frac{1}{4}}^{\frac{1}{2}} \int_0^{\frac{1}{2}} f(x, y) dx dy$$

$$\begin{aligned}
 \textcircled{4} \quad P\{(x, y) \mid 0 \leq x \leq \frac{1}{2}, \frac{1}{4} \leq y \leq \frac{1}{2}\} &= \int_0^{\frac{1}{2}} \int_{\frac{1}{4}}^{\frac{1}{2}} f(x, y) dy dx \\
 &= \int_0^{\frac{1}{2}} \int_{\frac{1}{4}}^{\frac{1}{2}} \frac{2}{5} [2x + 3y] dy dx \\
 &= \frac{2}{5} \int_0^{\frac{1}{2}} [2xy + 3y^2]_{\frac{1}{4}}^{\frac{1}{2}} dx \\
 &= \frac{2}{5} \int_0^{\frac{1}{2}} [x + \frac{3}{8} - \frac{x}{2} - \frac{3}{32}] dx \\
 &= \frac{2}{5} \int_0^{\frac{1}{2}} [\frac{x}{2} + \frac{9}{32}] dx \\
 &= \frac{2}{5} [\frac{x^2}{4} + \frac{9}{32}x]_0^{\frac{1}{2}} \\
 &= \frac{2}{5} [\frac{1}{16} + \frac{9}{64}] = \frac{13}{160}
 \end{aligned}$$

$$\frac{3}{8} - \frac{3}{32} = \frac{3}{8} (1 - \frac{1}{4}) = \frac{9}{32}$$

$$\begin{aligned}
 \textcircled{5} \quad \text{find } P(x \leq \frac{1}{2} \mid \frac{1}{4} \leq y \leq \frac{1}{2}) &= \frac{P(x \leq \frac{1}{2}, \frac{1}{4} \leq y \leq \frac{1}{2})}{P(\frac{1}{4} \leq y \leq \frac{1}{2})} \\
 P(\frac{1}{4} \leq y \leq \frac{1}{2}) &= \int_{\frac{1}{4}}^{\frac{1}{2}} h(y) dy = \int_{\frac{1}{4}}^{\frac{1}{2}} \frac{2}{5} (1 + 2y) dy \\
 &= \frac{2}{5} (y + 3y^2)_{\frac{1}{4}}^{\frac{1}{2}} \\
 &= \frac{2}{5} (\frac{1}{2} + \frac{3}{8} - \frac{1}{4} - \frac{3}{32})
 \end{aligned}$$

Example 26. The joint density for the random variables (X, Y) where X is the unit temperature change and Y is the proportion of spectrum shift that a certain atomic particle produces

$$f(x, y) = \begin{cases} 10xy^2 & 0 < x < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

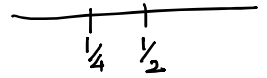
- Find the marginal distributions and conditional density $f(y|x)$
- Find the probability that the spectrum shifts more than half of the total observations, given that the temperature is increased by 0.25 unit

$$\begin{aligned}
 \textcircled{1} \quad \text{Marg. dist } g(x) &= \int_{-\infty}^{\infty} f(x, y) dy = \int_x^1 10xy^2 dy = 10x \left[\frac{y^3}{3} \right]_x^1 = \frac{10}{3} x (1 - x^3) \\
 \text{Marg dist } h(y) &= \int_{-\infty}^{\infty} f(x, y) dx = \int_0^y 10xy^2 dx = 10y^2 \left[\frac{x^2}{2} \right]_0^y = 10y^2 \frac{y^2}{2}
 \end{aligned}$$

$$\begin{aligned}
 f(y|x) &= \frac{f(x, y)}{g(x)} = \begin{cases} \frac{10xy^2}{\frac{10}{3}x(1-x^3)} & 0 < x < y < 1 \\ 0 & \text{otherwise} \end{cases} \\
 &= \begin{cases} \frac{3y^2}{1-x^3} & 0 < x < y < 1 \\ \dots & \dots \end{cases}
 \end{aligned}$$

$$f(x,y) = \begin{cases} 3y^2 / (1-x^3) & 0 < x < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$(2) \quad P(Y > 1/2 \mid X \geq 1/4) = \frac{P(Y > 1/2, X \geq 1/4)}{P(X \geq 1/4)}$$



$0 < x < y$

$$\begin{aligned} P(Y > 1/2, X \geq 1/4) &= \int_{1/2}^1 \int_{1/4}^y f(x,y) dx dy = \int_{1/2}^1 \int_{1/4}^y 10xy^2 dx dy \\ &= \int_{1/2}^1 [10x^2 y^2]_{1/4}^y dy \\ &= \int_{1/2}^1 [5y^4 - \frac{10}{16}y^2] dy \end{aligned}$$

$$P(X \geq 1/4) = \int_{1/4}^1 g(x) dx = \int_{1/4}^1 \frac{10}{3}x(1-x^3) dx =$$

Example 27. The joint PMF of (x, y) is given by $P(x, y) = k(2x+3y)$, $x = 0, 1, 2$; $y = 1, 2, 3$.

Find the marginal distribution

• conditional distribution of X given $Y = 1$

• conditional distribution of Y given $X = 2$

$$P(x, y) = k(2x + 3y)$$

$\begin{smallmatrix} x \\ y \end{smallmatrix}$	0	1	2	
1	3k	5k	7k	15k
2	6k	8k	10k	24k
3	9k	11k	13k	33k
	18k	24k	30k	72k

$$\sum_x \sum_y P(x=x, y=y) = 1$$

$$72k = 1 \Rightarrow k = \frac{1}{72}$$

(2)

Marg. dist on x

x	0	1	2
$g(x)$	18k	24k	30k

③ Marg. dist on y

y	1	2	3
$h(y)$	$15k$	$24k$	$33k$

④ cond. dist of x given $y=1$: $P(X=x|Y=1) = \frac{P(X=x, Y=1)}{P(Y=1)}$
 $= \frac{P(X=x, Y=1)}{h(1)}$

① $x=0$, $P(X=0|Y=1) = \frac{P(X=0, Y=1)}{h(1)}$
 $= \frac{3k}{15k} = \frac{1}{5}$

② $x=1$, $P(X=1|Y=1) = \frac{P(X=1, Y=1)}{h(1)} = \frac{5k}{15k} = \frac{1}{3}$

③ $x=2$, $P(Y=y|X=2)$?